

A Cyber Physical System for Tool Condition Monitoring using Electrical Power and a Mechanistic Model

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Abstract

Tool Condition Monitoring (TCM) systems, which aim to identify tool wear automatically to avoid damage to the machined part, are often not suitable for industrial applications due to: (i) sensing methods which can be expensive and invasive to install and (ii) testing regimes that only evaluate a narrow operating window under controlled conditions. To combat these issues, the research outlined in this paper explores the feasibility of TCM using electrical power consumption of an industrial Computer Numeric Control (CNC) cutting machine in combination with a mechanistic model for end milling operations. End milling of aluminium 6082 was performed over a range of cutting parameters (i.e. spindle speed, feed rate, tool diameter) and repeated with increasing levels of tool wear to ascertain the suitability of this approach. Reasonable correlation between the predicted and observed tool wear was found using the mechanistic model ($R^2 = 0.801$), however variance between the cutting parameters highlights the limitations of the predictions. To combat this variance, an averaging window is taken over the course of a cutting program to reduce the overall error. A concept TCM system is then proposed, utilising cyber-physical models of the milling process to determine the width and depth of cuts for complex geometries which change in real time, with the challenges of implementing such a system discussed.

Highlights

- Tool wear increased cutting power in experiments by 102-213%
- Mechanistic model provides a reasonable indicator for tool wear ($R^2 = 0.801$)
- Requires prior testing to establish coefficients and knowledge of cutting parameters
- Cyber Physical milling model outlined to determine cutting parameters from NC code

1. Introduction

Computer Numeric Control (CNC) has enabled the automation of machining processes, such as milling, turning and drilling, by programming machining operations and parameters such as paths, spindle speed and feed rates. However, CNC machines generally have limited or no knowledge of their working environment, meaning that they cannot detect, adapt or deviate from a cutting program to avoid issues such as chatter, tool wear and breakage, tool deflection or collisions without the intervention a human operator [1].

Cutting tools are consumable components of the CNC machine which degrade and wear with use through mechanisms such as abrasion, adhesion, diffusion, fatigue and chemical wear [2]. Tool wear can significantly impact the machining process of a part, reducing the quality of the surface finish and potentially damaging the part, resulting in either additional rework or the need to scrap the part [3]. Research into Tool Condition Monitoring (TCM) systems, where sensing, decision making and control strategies are introduced to detect the onset of tool wear, has therefore attracted a significant amount of interest from the academic community with a number of review articles listed [1][4][5][6][7][8]. However, key challenges remain, including the development of TCMs which can be cost effective to install and maintain and are robust and reliable in the presence of the variability and discontinuities of real-world machining.

Several sensing methods have been evaluated within the literature to determine tool wear indirectly including force [9][10], vibration [11], spindle power [12][13][14], acoustic emission [15], eddy current [16] or combinations of these (via sensor fusion) [17][18][19]. However, no research has been published focused on investigating the feasibility of monitoring the electrical power consumption of the entire CNC machine as a method of detecting tool wear. Electrical power monitoring has advantages over some of the aforementioned approaches due to its relative low cost [9], non-intrusive installation [7] and non-reliance upon open machine architectures to allow access and integration with existing sensor systems (e.g. spindle current sensors) [1], which may also cause interoperability and data veracity issues when deployed across different machines.

Developing robust methods of interpreting sensor data to infer tool wear levels causes additional challenges due to the variability and discontinuous nature of cutting processes such as end milling. These variabilities are caused by wide operating and transient cutting parameters, such as spindle speed, feed rate, depth and width of cut, tool type, workpiece geometry and material, discontinuous tool paths and noise from background operations influencing the measured signal (e.g. coolant pump, lighting, computers). Several black-box approaches, such as Artificial Intelligence (AI), have been used to identify tool wear with good accuracy [17][19][20]. However, these are usually only evaluated within limited operating conditions; therefore when changes occur to the system (i.e. different machines, tools,

workpiece materials) retraining of the algorithms is needed which requires further controlled testing and data collection. These approaches reduce the robustness and scalability of the system and increase the operating cost which will limit the relevance of the research to industry. On the other hand, white box approaches, such as mechanistic models, use scientific first principles to represent the real-world complexities through mathematical relationships, with examples developed to determine tool wear based upon changes in electrical spindle power [12] and force measurements [9]. Mechanistic models are useful as they can be reused over a number of scenarios by changing the model parameters to match the changes within the real world. However, it is difficult to model and measure all of the real-world complexities that may impact the tool wear measurements. This leads to an inevitable trade-off between accuracy and usability/robustness when developing models in real applications where not all parameters can accurately be predetermined.

Several studies relevant to tool wear condition monitoring are listed in Table 1. Shao et al [12] develop a mechanistic model to predict tool wear using the electrical power of the spindle and known cutting parameters. Whilst experimental results from this study appear to show good correlation with the mechanistic model, only limited validation of the technique is demonstrated over a small range of cutting parameters and tool wear is only presented as new or worn. Drouillet et al [14] present an interesting approach using neural networks to update and correct Remaining Useful Life (RUL) prediction of an end milling tool, using historical tool degradation patterns from the electrical spindle power. The approach is used to correct errors in RUL prediction which is caused by stochastic tool wear degradation. However, all predictions are based upon nominal spindle power (i.e. cutting power of a new tool), so at present the approach is unable to accommodate changing cutting parameters. Nouri et al [9] develop a real-time tool condition monitoring method to track changes within cutting force coefficients which are independent of cutting conditions and correlated with tool wear within a mechanistic model. A Cumulative Sum (CUSUM) control chart is used to determine the transition of the cutting coefficient from gradual wear to the failure region. Wang et al [21] use a combination of cutting force measurement and electrical power sensors to predict tool wear in drilling. A mechanistic model is used to predict the cutting force, whilst idle and auxiliary power demand of the motor is predicted based upon a regression model using empirical data.

Table 1 Comparison of studies related to tool condition monitoring

	Cutting Process	Monitoring Method	Tool Wear Model	RUL	Cutting Model	Additional Motor Losses	Real-time Implementation strategy	Number of different cutting conditions evaluated
Shao et al [12]	Face Milling	Electrical Power (Spindle)	Mechanistic Model	No	Mechanistic Model	No	Yes	5 (2x cutting teeth, 4 x depth of cut, 3x feed rate, 4 x cutting speed)
Drouillet et al [14]	End Milling	Electrical Power (Spindle)	No	Yes (neural network)	No	No	No	4 tools (each run with a different spindle speed)

Nouri et al [9]	End Milling	Force	Force Tracking Model	No	Mechanistic Model	N/A	Yes	5x experiments with tool (3x material, 5x spindle speed, 4 x radial immersion, 4 x axial immersion, 30+ feed rate)
Wang et al [21]	Drilling	Electrical Power and Cutting Force	Mechanistic Model	No	Mechanistic Model	Yes	No	12 (3x spindle speed, 4x feed rate)

This research investigates the use of electrical power monitoring of the entire CNC machine (rather than just the spindle) in combination with a mechanistic model as a low cost and process robust method to detect and determine tool wear in CNC end milling.

There are two novel aspects of this work: (i) a comprehensive assessment of the mechanistic model using only the electrical power under variable conditions (i.e. different spindle speed and feed rates), which has only been assessed under a limited set of experiments by Shao et al [12], and (ii) the proposal of a cyber-physical system to support parameter calculation of the mechanistic model from the digital part program to enable real time implementation of a TCM system.

The structure of the paper is as follows. In section 2 the mechanistic model used to determine cutting power and tool wear is discussed. Section 3 details the test methodology used to assess the mechanistic model under variable cutting conditions. In Section 4 the results of the experiments are presented and discussed, whilst within section 5 an approach to tool conditioning monitoring combining a mechanistic model and cyber-physical system is presented and discussed. Conclusions to the research and future work are presented in Section 6.

2. Mechanistic Model

A number of mechanistic models have been developed to predict cutting forces which are often expressed in terms of their tangential and radial components [9]. Electrical power monitoring, unlike force monitoring, does not detect individual axial components. Rather the magnitude of the total force is inferred through characterisation of the electrical power signal. Within this work the suitability of a mechanistic model originally developed by Shao et al [12] is assessed which is focused upon the tangential force elements which can be detected within the spindle electrical power signature. In Shao et al's approach the average cutting force can be modelled using equation (1).

$$F_c = K\bar{h}^{-c}a_p a_e \quad (1)$$

Where F_c is the average cutting force, K is the cutting force constant for the workpiece material (N/mm^2), $\bar{h}$ is the average chip thickness (mm), c is the workpiece tool coefficient, a_p is the depth of cut (mm) and a_e is the width of cut (mm). Shao et al [12] calculate $\bar{h}$ is using equation (2).

$$\bar{h} = \frac{f_z}{\psi} (\sin\psi_1 + \sin\psi_2) \quad (2)$$

Where f_z is the feed per tooth (mm) which is calculated using equation (3), ψ is the immersion cutting angle (rad) calculated using equation (4), and ψ_1 and ψ_2 are the parts of the immersion angle (rad), shown in Figure 1 and calculated in equations (5) and (6).

$$f_z = f/Zn \quad (3)$$

$$\psi = \psi_1 + \psi_2 \quad (4)$$

$$\psi_1 = \arcsin \left[\frac{a_e + 2\delta}{D} \right] \quad (5)$$

$$\psi_2 = \arcsin \left[\frac{a_e - 2\delta}{D} \right] \quad (6)$$

In equations 3-6, f is the feed rate (mm/min), Z is the number of cutting teeth on the tool, n is the spindle speed (rpm), δ is the offset between the centre of the cutting tool (mm) and the cutting width (a_e) and D is the diameter of the cutting tool (mm). Cutting power ($P_{cutting}$) can then be determined by multiplying the force by the feed rate, shown in equation (7):

$$P_{cutting} = F_t f = f K \bar{h}^{-c} a_p a_e \quad (7)$$

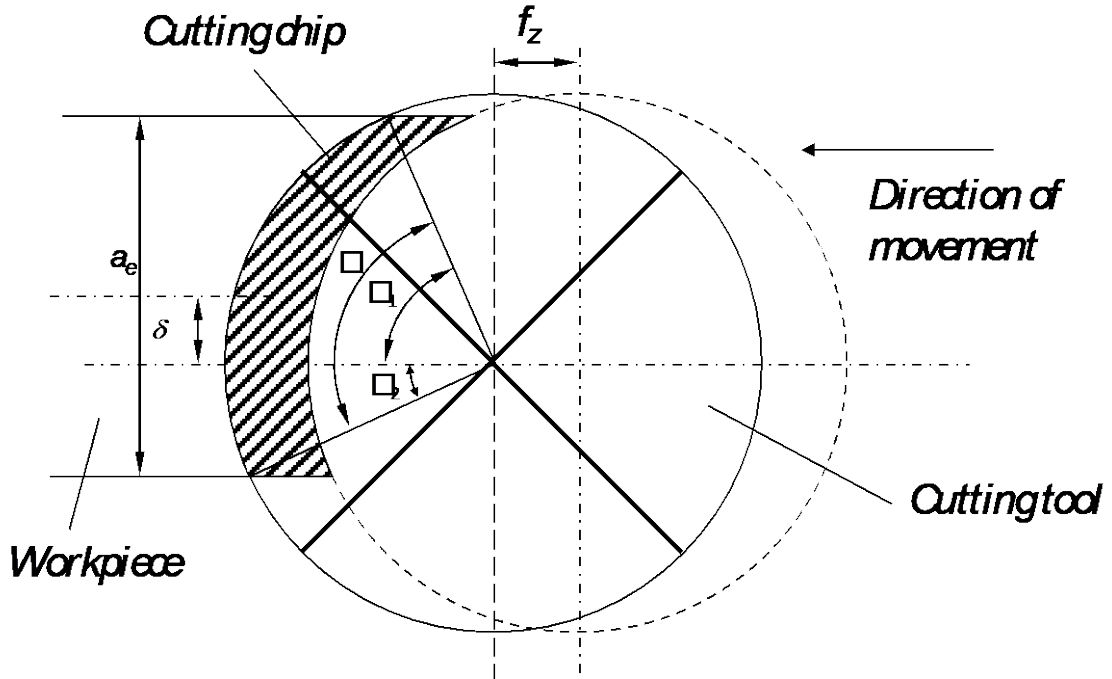


Figure 1 Schematic of the cutting parameters used within the mechanistic model adapted from [12]

Flank wear along the tool edge causes additional friction during cutting, increasing the overall power required to perform a cut shown in Figure 2. The effect of tool wear upon the cutting power consumption is modelled below using equations (8) and (9).

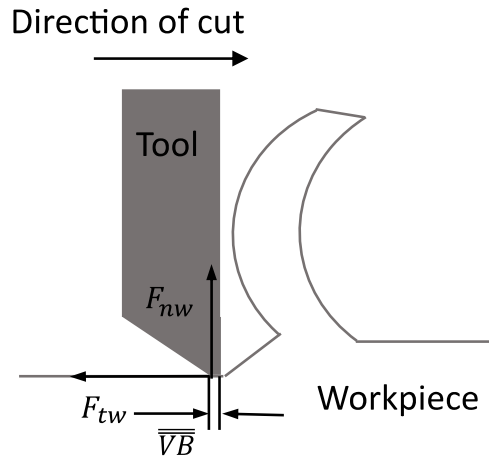


Figure 2 Impact of flank wear ($\overline{VB}$) upon the normal and tangential cutting forces

$$F_{nw} = gH\overline{VB}a_p \quad (8)$$

$$F_{tw} = \mu F_{nw} = \mu gH\overline{VB}a_p \quad (9)$$

Where F_{nw} is the normal force due to wear, F_{tw} is the tangential force due to wear, $\overline{VB}$ is the average flank wear (mm), H is the Brinell hardness of the material and g is gravity (Note: H is multiplied by g to convert from kgf/mm to N/mm) and μ is the coefficient of friction between the workpiece and the cutting tool. Combining these forces gives a total due to wear which need to be overcome by the spindle and feed motors, shown in equation (10).

$$F_{RW} = F_{nw} + F_{tw} = (1 + \mu)gH\overline{VB}a_p \quad (10)$$

The additional power to overcome the increased friction due to tool wear (P_{wear}) can then be determined using equation (11).

$$P_{wear} = Z \frac{n}{120\pi} \frac{D}{2} a_p [(1 + \mu)gH\overline{VB}\psi] \quad (11)$$

Rearranging equation (11) enables $\overline{VB}$ to be determined when P_{wear} is known, shown in (12).

$$\overline{VB} = \frac{240\pi P_{wear}}{ZnDa_p(1 + \mu)gH\psi} \quad (12)$$

3. Methodology

3.1. Experimental Setup

Experiments have been designed to investigate if the amount of tool wear can be determined based upon the measured electrical power under multiple cutting parameters using the test procedure shown in Figure 3.

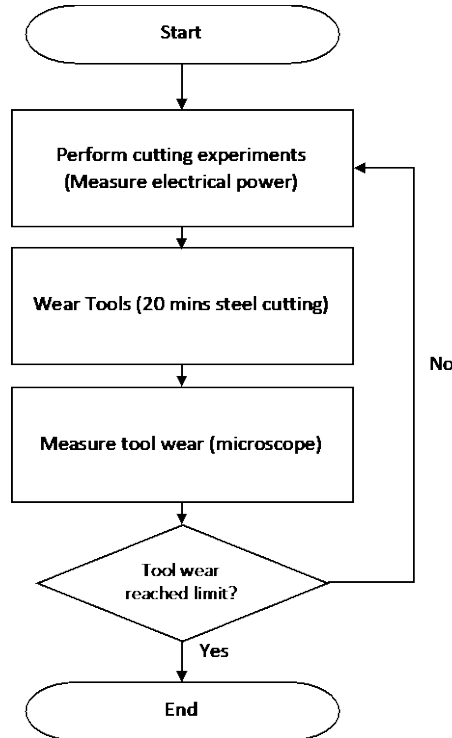


Figure 3 Test Procedure

All cutting experiments were conducted using a Hurco 3-axis VM1 CNC machine. Cutting tools chosen for the experiment were four flute High Speed Steel with 8% Cobalt (HSS-Co8) end mills with three different diameters (6mm, 8mm, 10mm) to examine the effect of tool wear for different tool diameters. Aluminium grade 6082 T651 was used for the workpiece material. Straight line cutting was performed to gather consistent measurements, with a

depth of cut and width of cut being 1x and ¼x (tool diameter) respectively for each cutting tool used. Cutting parameters were adjusted based upon two parameters (torque, cutting speed) each consisting of 3 levels (nominal (i.e. determined from machine tool data sheet), nominal +10%, nominal -10%), resulting in nine unique cutting experiments per tool (i.e. 27 in total) which are shown in Table 2. Torque and cutting speed were chosen rather than spindle speed and feed rate to ensure sensible values were used which may otherwise have damaged the tool or workpiece due to excessive torque (i.e. in the cases when low spindle speed and high feed rate could have been combined). Cutting torque can be determined from the cutting power as shown in equation (13).

$$T = \frac{30P_c}{\pi n} \quad (13)$$

Where T is the torque (Nm), P_c is the cutting power (W) and n is the spindle speed (rpm).

Table 2 Cutting Parameters for each tool

Tool Diameter (mm)	Spindle Speed (rpm)	Feed Rate (mm/min)	Estimated Torque (Nm)	Material Removal Rate (cm ³ /min)
6	5600	470	0.16	4.2
6	6700	565	0.16	5.1
6	4400	371	0.16	3.3
6	4625	470	0.18	4.2
6	6900	470	0.13	4.2
6	5600	565	0.18	5.1
6	8300	565	0.13	5.1
6	5450	371	0.13	3.3
6	3650	371	0.18	3.3
8	4000	580	0.42	9.3
8	4800	696	0.42	11.1
8	3200	464	0.42	7.4
8	3310	580	0.48	9.3
8	4945	580	0.36	9.3
8	3970	696	0.48	11.1
8	5935	696	0.36	11.1
8	2650	464	0.48	7.4
8	3950	464	0.36	7.4
10	3100	600	0.81	15.0
10	3720	720	0.81	18.0
10	2480	480	0.81	12.0
10	2560	600	0.93	15.0
10	3830	600	0.69	15.0
10	3070	720	0.94	18.0
10	4600	720	0.69	18.0
10	2050	480	0.93	12.0
10	3060	480	0.69	12.0

Three phase current and voltage measurements were taken for the entire machine at a sample frequency of 50 kHz using National Instruments NI-9242 and NI-9247 within cDAQ-9174 chassis, with the experimental set up shown in Figure 4. Root Mean Squared (RMS) analysis was conducted to determine active power at a resolution of 5Hz.

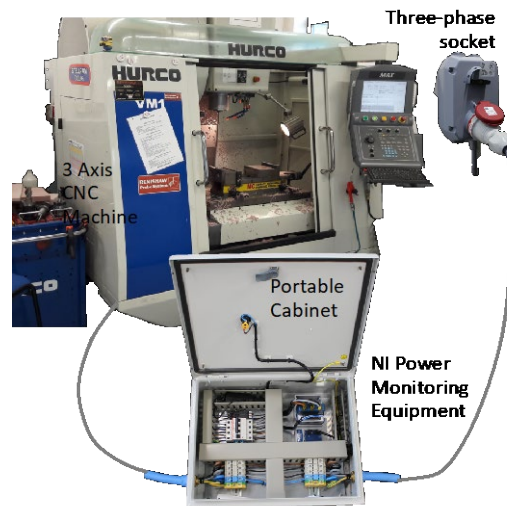


Figure 4 Experimental setup with power monitor connected between CNC machine and plug socket

Tools were worn for 20 minutes intervals using EN8 grade alloyed carbon steel (201-225 Brinell hardness) and constant cutting parameters to accelerate wear. Tools were removed from the CNC machine and measured under a microscope to determine values for localised and uniform flank wear in accordance with ISO 8688-2:1989 [22], as shown in Figure 5. *Localised wear* is the maximum local wear region, usually found at the top of the mill flute (see area highlighted in Figure 4). *Uniform wear* is identified by a region of wear which remains largely constant down the length of the cutting edge. Measurements are taken along this region and averaged to determine the uniform wear value. This process was repeated until a wear threshold of either 0.5mm (localised) or 0.2mm (uniform) was exceeded at which point the experiments involving that tool were concluded.

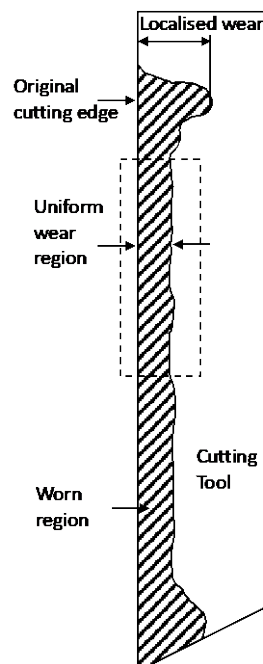


Figure 5 Typical flank wear profile for end milling tool, adapted from ISO 8688-2:1989 [22]

3.2. Data Analysis

The cutting power for each of the experiments was determined by taking an average over the three cutting passes, omitting the first and last second of data to avoid unwanted features caused by material break in or out, highlighted in Figure 6. The change in power due to tool wear P_{worn} can then be determined by subtracting the total active power for a particular cutting experiment involving a new tool, P_{total_new} from total electrical power P_{total_worn} for each worn tool experiment, shown in equation (14). The cutting power $P_{cutting}$ for a tool was determined using equation (15) by subtracting the power required to maintain the spindle and background auxiliary power $P_{spindle}$ from the total measured power P_{total} . $P_{spindle}$ was determined by performing the same experiments without the workpiece material (i.e. air cutting).

$$P_{worn} = P_{total_worn} - P_{total_new} \quad (14)$$

$$P_{cutting} = P_{total} - P_{spindle} \quad (15)$$

Cutting coefficients K , HB , μ and c need to be obtained for use in equations (12). Values for K and c have been obtained from Sandvik Coromant [23] as 400 N/mm² and 0.25 respectively. HB was determined using a Brinell hardness test with a value of 76.25 HB obtained, whilst μ was determined to be 0.18 using least squares analysis from the data set collected.

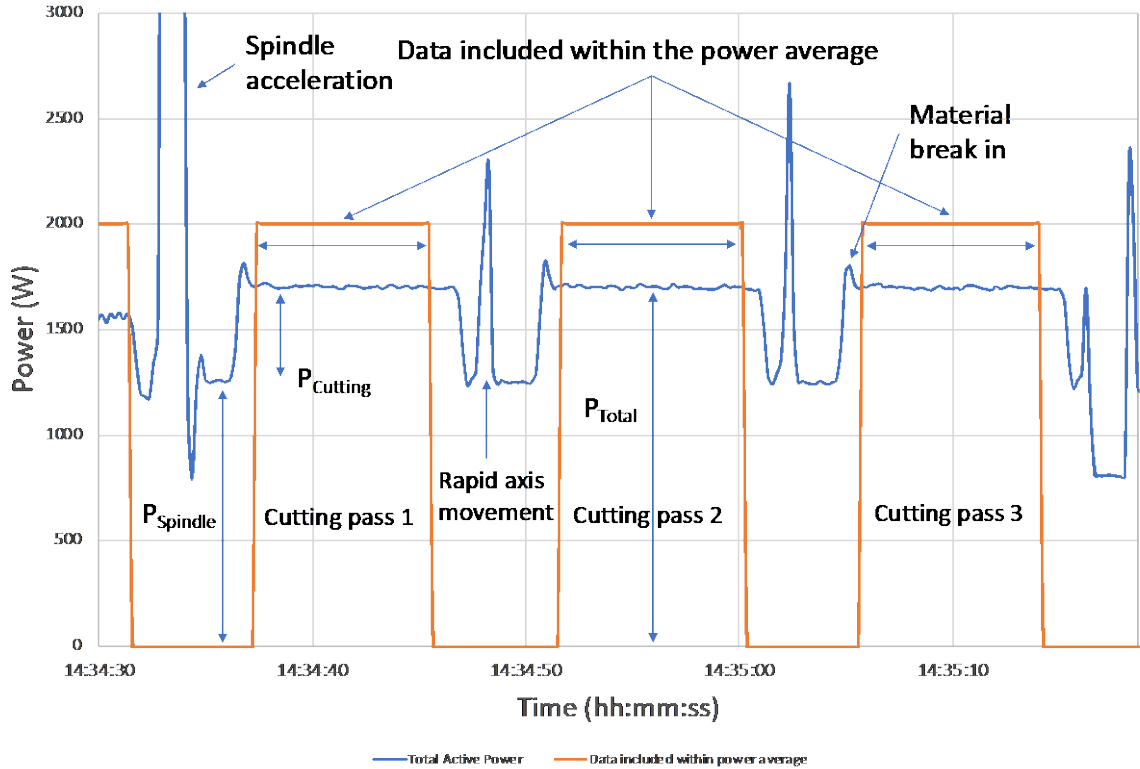


Figure 6 Power curve showing 3 passes of the same cutting parameter with the areas selected of average

4. Experiment Results and Discussion

4.1. Prediction of tool wear

The first part of the analysis explores the suitability of the electrical power monitoring of the CNC machine and the mechanistic model outlined in equation (12) to predict the amount of tool wear. The experiment was conducted in accordance with the methodology shown in Figure 4, with a total of 8 different levels of wear per tool. The tools showed progressive wear, highlighted in Figure 7 and Figure 8, with the 6mm exceeding the recommended limits of 0.2mm uniform wear after 140 minutes of cutting on steel and the 10mm also approaching this limit (i.e. 0.171mm). It is noted that the 8mm tool however did not show the same level of wear within the same time period (i.e. 0.138mm), potentially highlighting the variability in the rate at which tools wear.

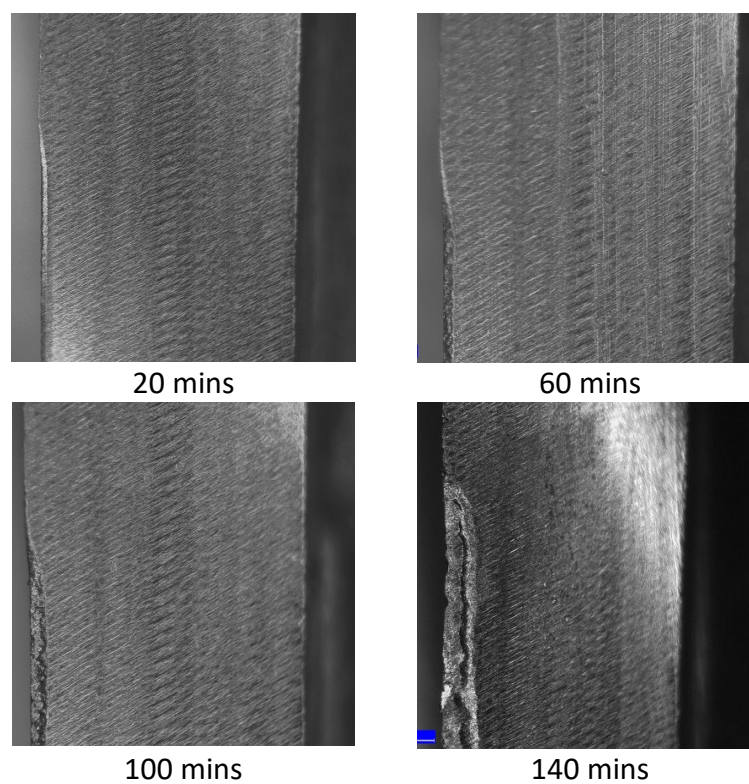
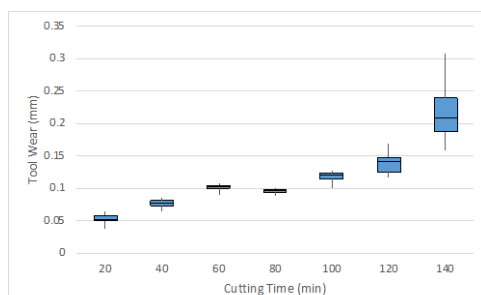
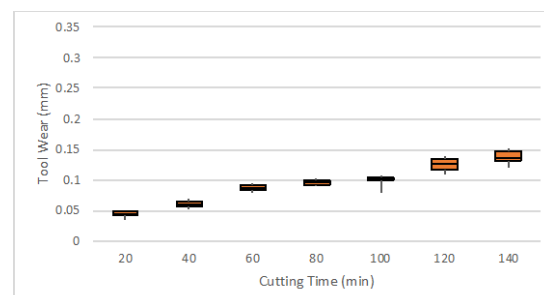


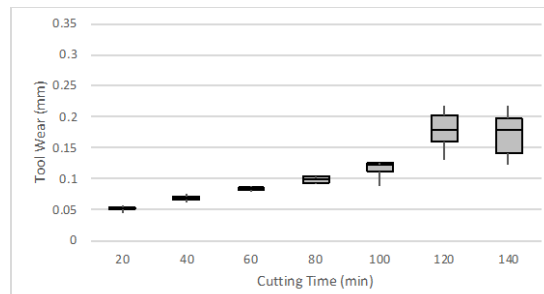
Figure 7 Progression of tool wear on the flank edge for the 10mm tool



(a)



(b)



(c)

Figure 8 Flank wear progression shown on box and whisker plots measured on the microscope for end mill diameters of (a) 6mm, (b) 8mm and (c) 10mm

Electrical (cutting) power increased with tool wear, as shown for the nominal cutting parameters for each tool size in Figure 9 and detailed for the various cutting parameters used with the 6mm tool in Figure 10. When the 6mm tool was in its most heavily worn state (i.e. after 140 minutes of cutting), the cutting power increased, relative to new, by 113-350W depending upon cutting parameters, equating to an 102-213% increase in cutting power and 18-31% increase in total power of the CNC machine (i.e. including spindle and auxiliary power consumption).

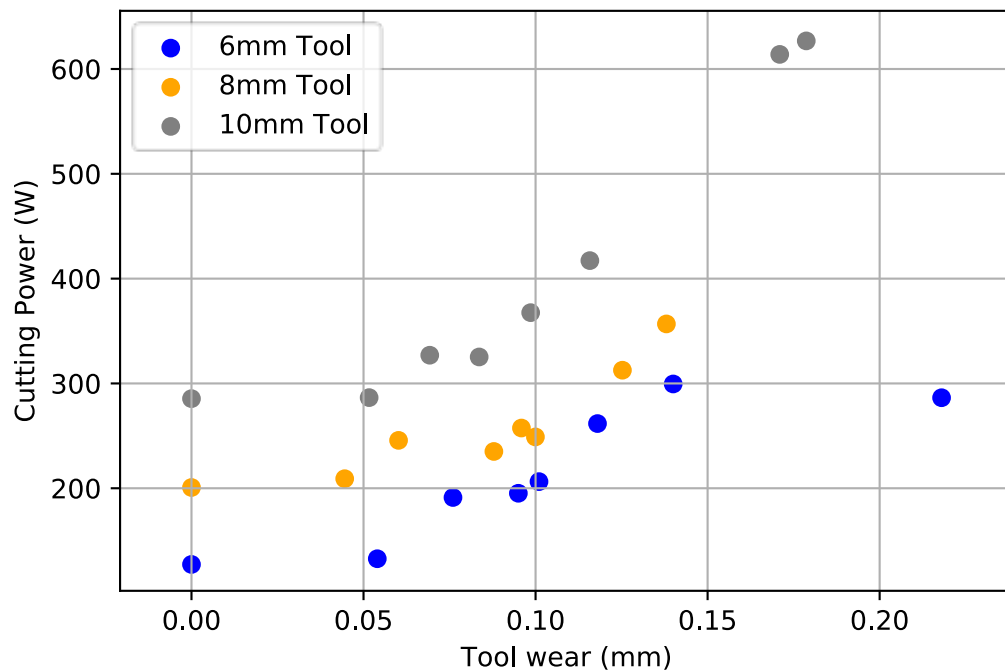


Figure 9 Cutting power vs tool wear for the nominal cutting parameters (spindle speed and feed rate) of each tool (6mm – 5600rpm, 470 mm/min, 8mm- 4000rpm, 580 mm/min, 10mm- 3100rpm, 600 mm/min)

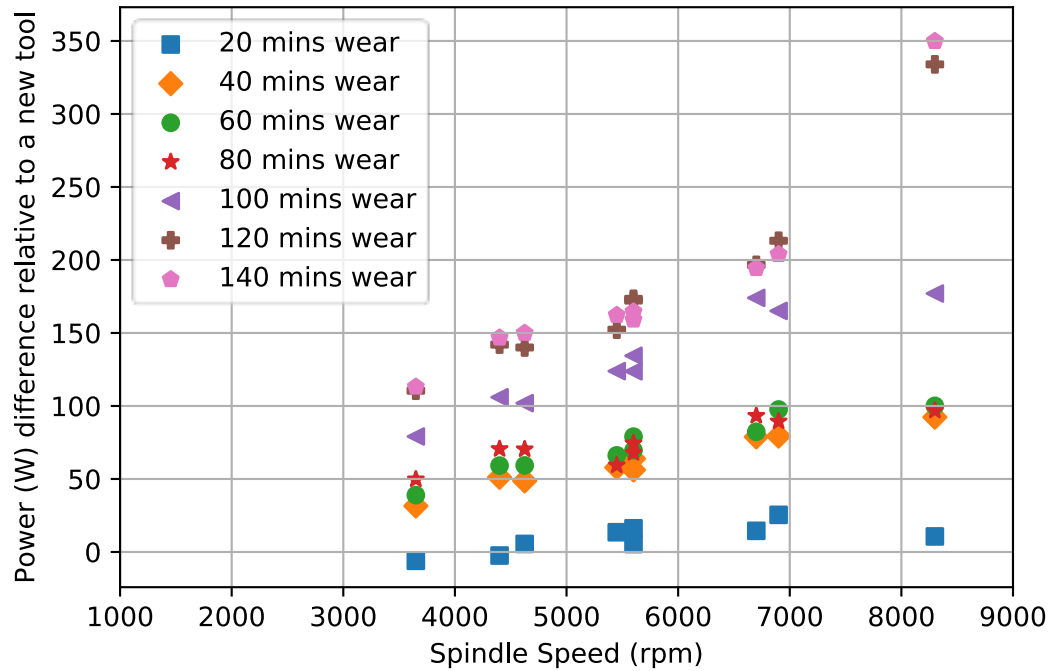


Figure 10 Change of cutting power with different spindle speeds and tool wear for cuts conducted with the 6mm tool

Tool wear is predicted using the mechanistic model presented in equation (12) for every experiment conducted using the electrical power data. Each prediction is then compared to the observed tool wear measured using the microscope and is shown in Figure 11. Reasonable correlation between the predictive and observed tool wear was found ($R^2 = 0.801$). During the early stages of wear (i.e. less than 0.1 mm) there is a tendency for the mechanistic model to underestimate or even report a negative wear value; a phenomenon also reported in other literature [14] [21]. It should be noted that the clusters of nine vertical points in Figure 11 correspond to the nine cutting parameters experiments used for each tool, whilst their appearance on a single vertical line is due to the same microscope reading (which is taken after all nine cuts are complete but before further wear is induced).

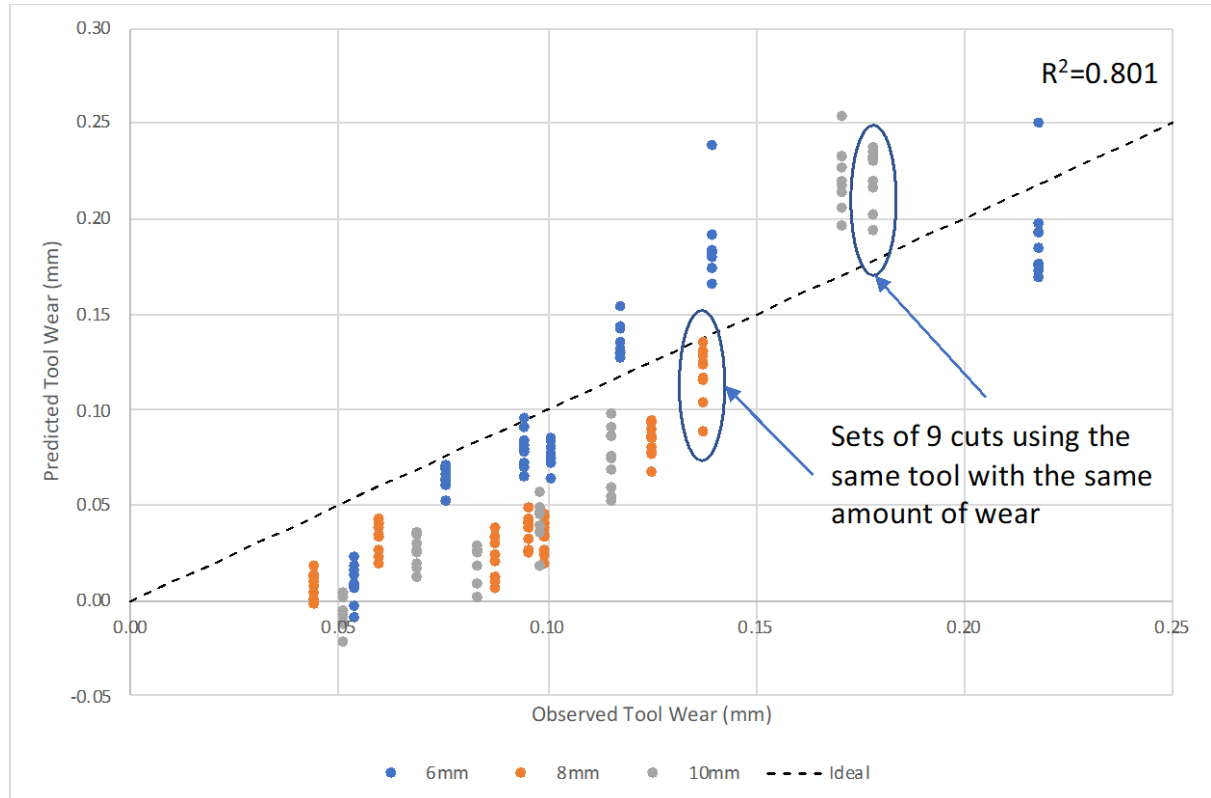


Figure 11 Actual tool wear vs predicted tool wear using electrical power and equation (12)

Whilst the general trend is positive, there is noticeable variance between the observed and predicted wear, which will impact the accuracy of the prediction and reduce the confidence in the TCM. To try and understand the source of the variance, potential factors are explored below.

Some of the variance can be attributed to errors in the observed tool wear measurement (i.e. microscope value) as can be seen in Figure 8. Optical measurement of end mill tool wear is difficult to quantify precisely due to its variability along the flank edge, whilst the helix geometry of the tool can make the alignment difficult and cause shadows within the image [18], as can be seen in Figure 7. These effects become more pronounced as the tool wear increases, highlighted in the increased variance of the worn tools in Figure 8. This will therefore have the impact of shifting the nine experiments conducted with the same tool at a particular wear level either left or right along the x axis. Errors from the electrical power sensors were much lower (average error in the mean were less than 1W which equates to approximately 0.7×10^{-3} mm of wear within the mechanistic model) due to the large number of samples and their relative consistency (see Figure 6) and thus can be ignored as a significant source in variance.

For cases where the tool wear is constant across experiments but cutting parameters change (i.e. 9 cutting parameters for each stage of tool wear for each tool), variance can be clearly seen in Figure 11, with a standard deviation of 0.012mm of tool wear. This value decreases to

0.077mm when the tool wear is low (i.e. less than 0.1mm of wear) but increases to 0.017mm for higher tool wear (i.e. above 0.1mm of wear).

To explore if this variance within the predicted wear is caused by a particular parameter or the order of cutting, the difference to the mean in tool wear between the individual prediction and the average for the set has been plotted against spindle speed, torque and cut number; as shown in Figure 12. Interestingly no pattern is evident that correlates the variance to any of parameters, concluding that the variance must result from either an uncontrolled, unobserved parameter or random fluctuation in measurements. This makes it difficult to determine the amount of tool wear precisely directly from electrical power signal. However, fuzzy values such as low, medium and high could be utilised in future work as a method of providing approximate feedback on tool wear, as was used by Cuka and Kim [18].

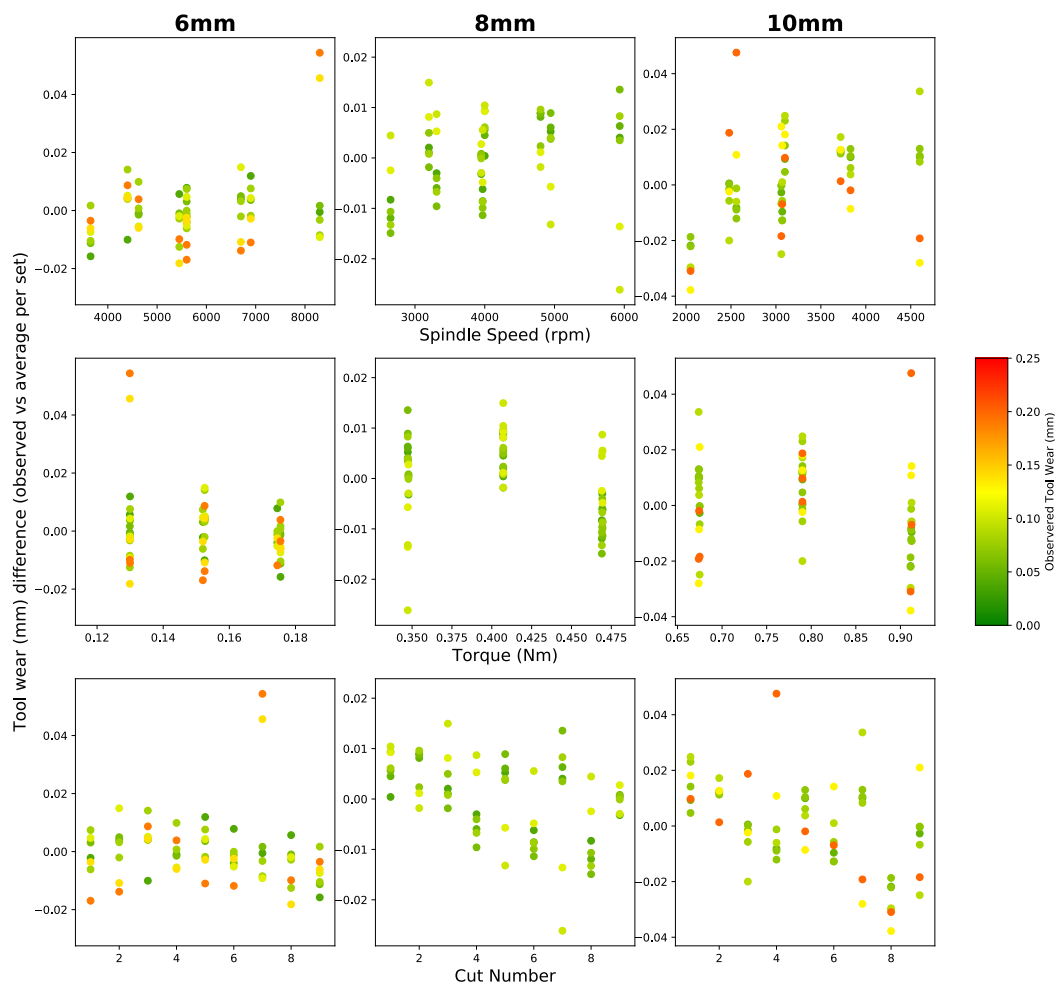


Figure 12 Variation within the tool wear predictions based upon the electrical power signals for each tool size

4.2. Prediction of cutting power using the mechanistic model (No tool wear)

The suitability of equation (7) to predict the cutting power when no tool wear is present is assessed in the following. Determining a value for the cutting power of a new tool P_{total_new} is vital to enable tool wear to be predicted using equation (12) and (14). Using a mechanistic model to predict cutting power with a new tool would enable greater flexibility for any TCM system, particularly for low quantity production or one-off jobs (typical within the defence, aerospace and low volume/prototyping automotive domains), rather than the alternative time-consuming approach of recording the power for a particular part program through a calibration test.

Very good correlation ($R^2=0.99$) is found between the predicted and observed cutting power, shown in Figure 13, although the model has consistently underpredicted the observed values by an average of 39.7 W. Further exploration of the data is shown in Figure 14, where the difference between the observed power and the predicted power (from equation (7)) plotted against the spindle speed and torque (where the torque is calculated using equation (13)). The difference between the observed and predicted cutting power is between 30-50 W for all cuts shown in Figure 14 (a) and (b). However, when differences between the observed and predicted are expressed as a percentage of the observed cutting power, to account for the differing cutting powers required for the experimental parameters (i.e. 6mm diameter tools require less power than 10mm due to less material removal), there is a significant difference of between 10-65% is identified, shown in Figure 14 (c) and (d). Interestingly a logarithmic trend is evident when comparing the percentage difference between the measured and predicted cutting power with torque in Figure 14 (d), with lower torque values (as seen with the 6mm end mill) resulting in much larger difference between the measured and predicted power. As a result, the prediction is much less accurate for cuts requiring a lower torque, which has a significant impact upon the suitability of using this approach for these parameters as the increase in absolute cutting power for worn tools is more subtle. Additionally, the spindle speed also appears to impact the accuracy, with lower spindle speeds resulting in a worse prediction for a given cutting torque, again highlighted in Figure 14(d).

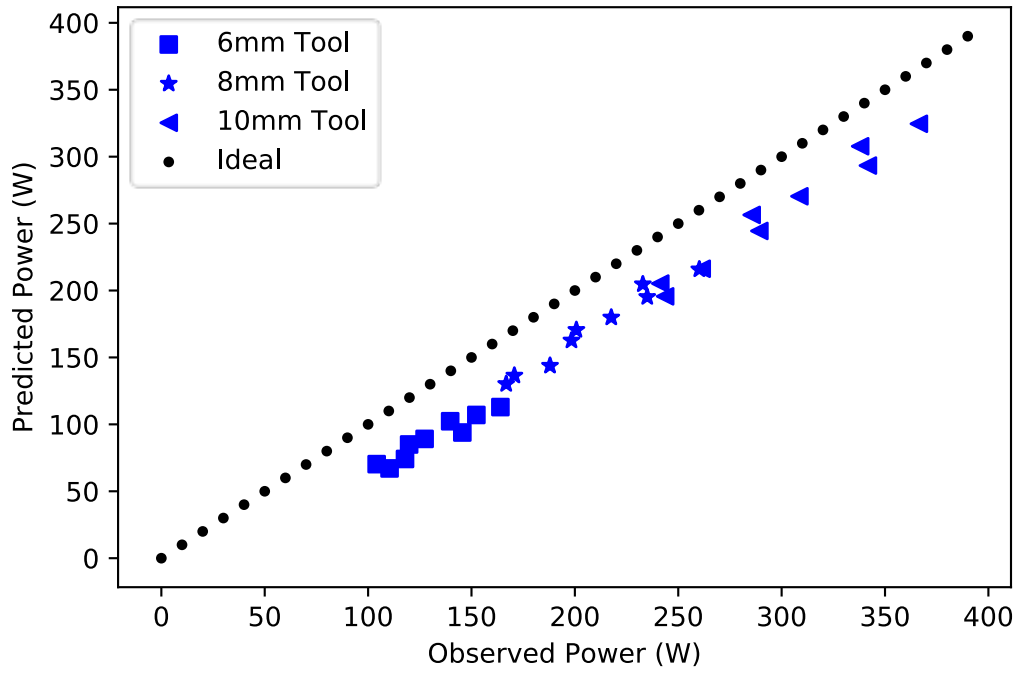


Figure 13 Observed vs predicted cutting power (W) for a tool when no wear is present

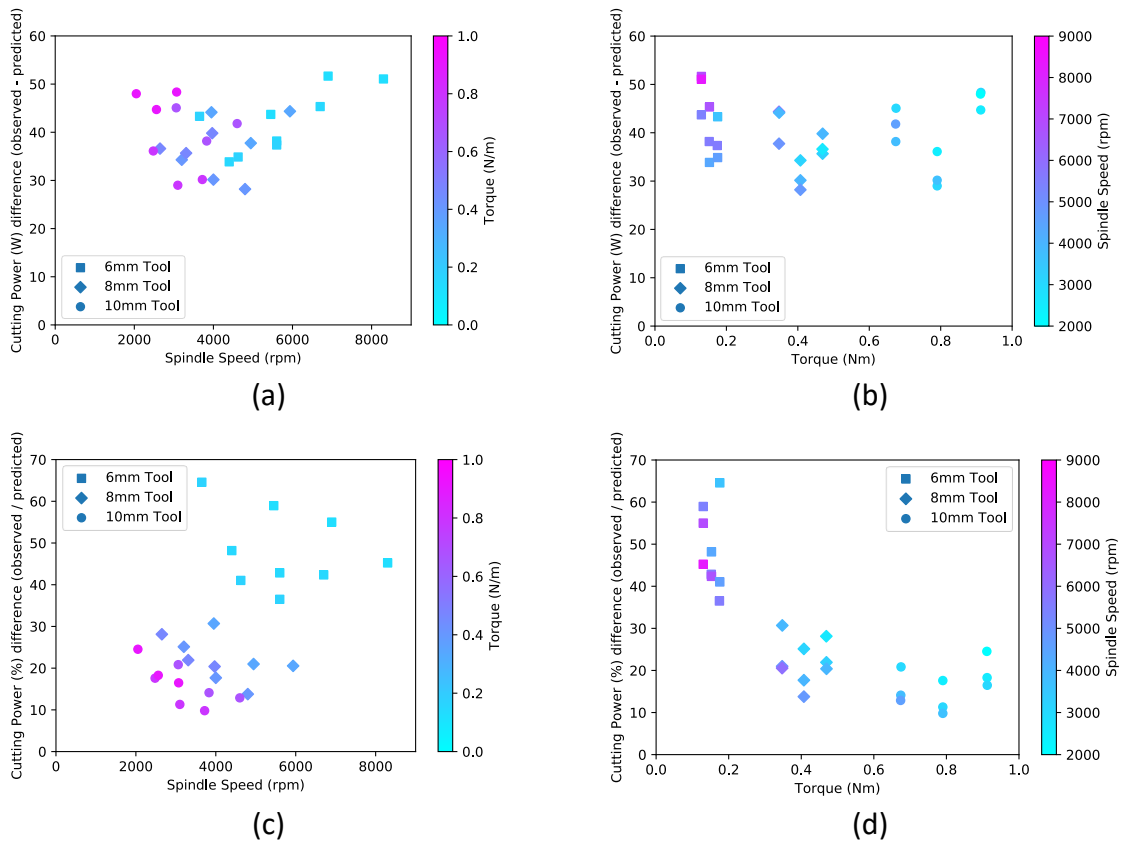


Figure 14 Absolute residual cutting power (W) (i.e. difference between observed and predicted cutting power) vs (a) spindle speed (rpm) and (b) torque (Nm), then the normalised residual power (%) (i.e. percentage difference between measured and predicted cutting power) vs (c) spindle speed (rpm) and (d) torque (Nm)

This relationship could be the result of motor efficiency variability, with lighter loads (torque) resulting in lower efficiency [24]. The potential variation in motor efficiency therefore complicates the cutting power prediction as quantification would require further data to be collected to map the motor efficiency for each individual CNC machine. This is further complicated as the efficiency of electric motors is known to vary based upon the machine's condition and environment (i.e. temperature, humidity), whilst these effects are compounded when the motor load is considerably less than the rated motor load [25].

5. Online Tool Monitoring

5.1. Challenges

The previous section has demonstrated that the level of wear of end milling tools can be estimated using a mechanistic model and electrical power consumption data. Within this section the concept for a TCM system is proposed combining the mechanistic model with a cyber-physical model of the milling process.

Unlike the straight-line cuts performed in section 4, real world milling is used to cut more complex geometries, where the cutting parameters (i.e. spindle speed, feed rate, width and depth of cut) will vary throughout the cutting program, inherently affecting the power consumption. An example of pocket milling can be seen in Figure 15, with changing width and depth of cut causing variability within the cutting power highlighted in Figure 15 (b). To utilise the mechanistic model evaluated within section 4, TCM services need to be able to determine the cutting parameters dynamically, particularly the depth and width of cut. Whilst the Numeric Control (NC) program provides information about the tool path, spindle speed and feed rate, it doesn't provide information on depth and width of cuts. These parameters will need to be determined via additional information including tool and workpiece geometry, previous material removal, as well as material hardness and friction coefficients. A cyber-physical model of the milling process is therefore proposed to support the determination of the width and depth of cuts in real time.

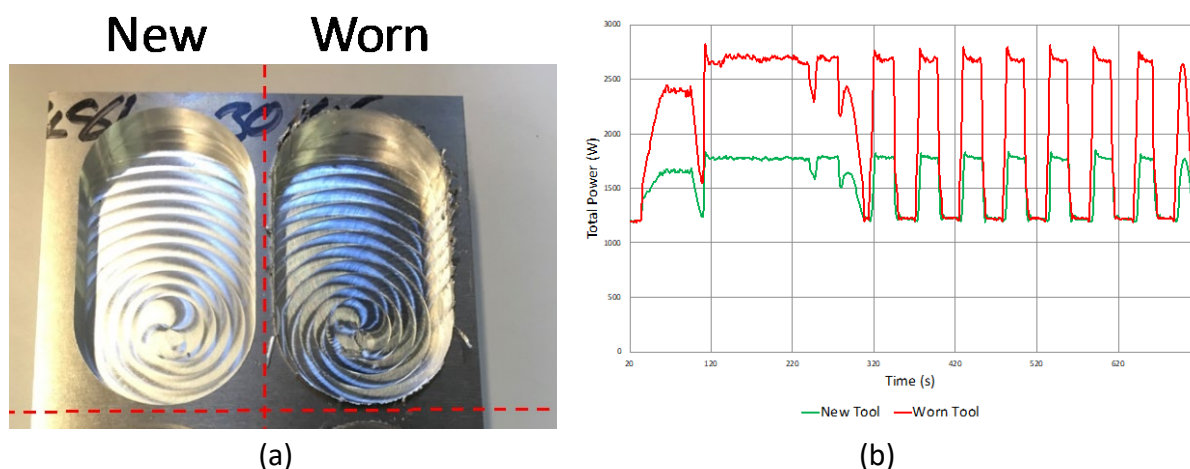


Figure 15 Pocket milling demonstrating varying width and depth of cuts with the difference in power for a new and worn tool

5.2. TCM Concept

To address the above challenges, a concept TCM is proposed, with three key components: (i) a virtual milling model, (ii) mechanistic model and (iii) a moving average filter, shown within a schematic diagram in Figure 16. The virtual milling model is used to determine if the CNC machine is cutting at a given instance in time within the part program and, if so, identify the volumetric cutting parameters (i.e. width and depth of cut). Interaction between the cutting tool and the workpiece is determined by simulating the tool path based upon instructions extracted from the NC code, shown in Figure 17. When interaction is detected volumetric cutting parameters are determined and removed from the workpiece model. The volumetric

cutting information is then passed to the mechanistic model to predict tool wear, along with the spindle speed (also from the NC code), the Brinell Hardness of the workpiece and coefficient of friction (determined from experiments beforehand). P_{wear} is calculated by subtracting P_{total_new} from P_{total} (see equation (14)) which is measured directly from the electrical power monitor. P_{total_new} can either be obtained from a calibration test of the part program (such as when proving out the part) or from a prediction using the mechanistic model for cutting power (equation (7)) combined with the virtual model of the milling process to obtain the relevant cutting parameter inputs. This enables an instantaneous tool wear prediction to be made. Finally, a moving average filter is used to account for the variability within the tool wear prediction identified in section 4.1. A simple moving average filter is chosen, rather than a weighted filter due to the progression of wear being slow relative to the average window. An alarm is triggered once the averaged wear exceeds the threshold of 0.2mm.

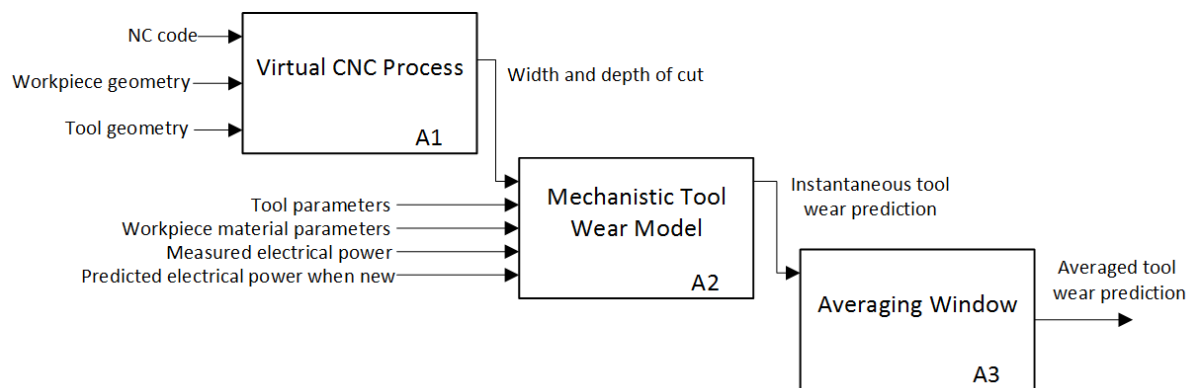


Figure 16 IDEF0 representation of the proposed TCM system

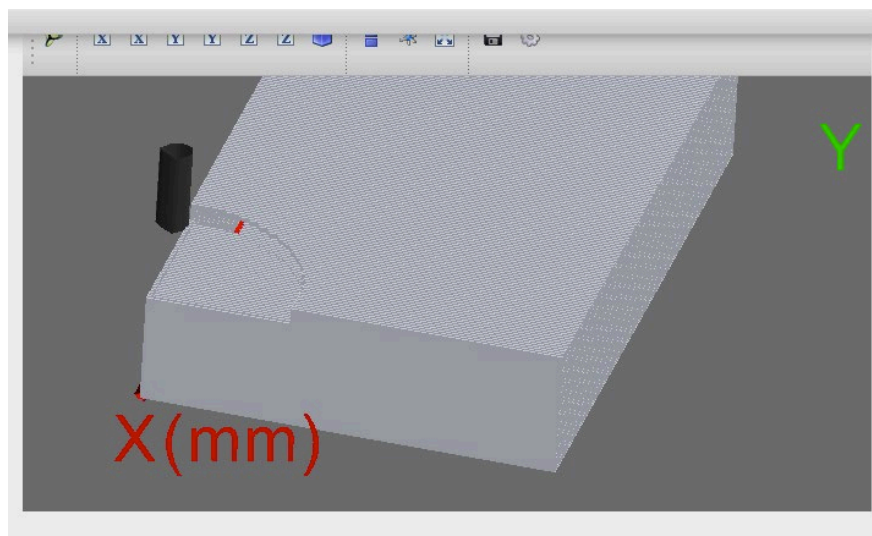


Figure 17 Virtual CNC model to determine when cutting occurs and determine volumetric cutting parameters

5.3. Discussion, limitations and future work

Developing and evaluating the performance of the TCM concept is beyond the scope of this paper, however a number of limitations and challenges have been outlined as useful avenues for future work.

Synchronising the virtual model with the real-world – Stochastic factors within the milling process can cause the virtual model to drift out of phase with the real-world milling process (e.g. the time to accelerate or decelerate the spindle) resulting in errors calculating tool wear. Mechanisms need to be developed to identify these drifts and feedback to correct the virtual model in real-time. Potential mechanisms could include feature detection (i.e. spindle acceleration and material cutting) to align signals by employing techniques such as Hidden Markov Models [26].

Data sampling and process selection – The ability to utilise electrical power measurements with consistent operating parameters (i.e. width and depth of cut) is greatly reduced in real world industrial milling, highlighted with the pocket milling example in Figure 15. This problem intensifies with increased speed of milling due to the shorter time required to perform cutting typical of industrial operation. Considerations therefore need to be made regarding the data sampling period and sampling rules to mitigate the effect of material break in/out, as shown in Figure 6.

Efficient system calibration and learning – The cost and time of calibrating and maintaining a TCM system is an important consideration when assessing the viability of a production TCM. If these costs become too high, then the value of a TCM system to a business is reduced. Our approach requires a number of parameters to be calibrated prior to on-line use (i.e. friction and material hardness), for every workpiece material and tool combination. Future research should be focused upon developing suitable techniques to reduce the number of off-line calibration experiments required whilst incorporating on-line feedback mechanisms to update models, such as cross referencing with other data sources (i.e. surface finish of parts from quality assurance tests) to infer the validity of the model.

Incorporation of auxiliary power – The effect of motor efficiency hasn't been incorporated within the power calculations. Whilst the efficiency is high at loads close to the motor rating (>90%), under lighter loads the efficiency can drop significantly. Developing an efficiency mapping of the motor at different spindle speeds and torque levels by combining the power signal with additional force sensors, such as Wang et al [21], may provide greater accuracy within the prediction.

Prediction of low power machining - Challenges still remain in predicting tool wear within low power machining (i.e. small volume, low feed rate, small tool) and with low cutting speeds. These are due to the relative low proportion of power for cutting compared with power

required for the spindle and other auxiliary functions which is particularly an issue in large machines.

Remaining Useful Life Prediction (RUL) – The mechanistic model is used to predict the current state of tool wear, however a more useful metric would predict RUL of the tool to enable users to consider if it requires changing before commencing the cutting program. RUL requires greater understanding of tool degradation patterns and how it is affected by cutting parameters and workpiece material. Specific challenges for developing RUL predictions align with *system calibration and learning* such as how to obtain reliable and cost effective prediction data which can be highly uncertain [27]. Whilst RUL is beyond the scope of this paper it should be considered in future research. Incorporating our approach with Drouillet et al [14] would be a useful research avenue.

6. Conclusion

This research has investigated the use of electrical power monitoring of a CNC machine as a method of detecting tool wear. A mechanistic model is used to predict the level of tool wear based upon cutting parameters and the difference between measured electrical power for new and worn tools. Experiments were conducted to evaluate the suitability of electrical power monitoring and the mechanistic model to detect tool wear and predict cutting power under various conditions (spindle speed, feed rate and tool sizes). Key findings include:

- Electrical power increased significantly with tool wear (102-213% of the cutting power which represents 18-31% increase in total CNC power).
- Reasonable agreement with the mechanistic model prediction and tool wear measured using microscope imaging measurements ($R^2 = 0.801$). However, variation within the predicted tool wear was observed between the cutting parameters (standard deviation 0.0139mm) suggesting the model is more suitable as an indicator for tool wear rather than a precise measurement.
- The cutting power mechanistic model for a new tool again provides good correlation with the observed data, however, is offset by between 30-50 W throughout. This is significant in the prediction of tool wear, particularly for small cutting powers (i.e. lower material removal and torque) where the impact of tool wear upon increased electrical power will be lower, resulting in greater error in the tool wear prediction.

Based upon these findings a concept for a Tool Condition Monitoring (TCM) system has been proposed which utilises the mechanistic model and a virtual milling model to determine tool wear condition based upon differing cutting conditions within the part program, thus supporting the implementation within real world cutting programs. There are however a number of limitations to the study which present opportunities for future work. These include synchronising the virtual model with the real-world, efficient system calibration and learning, incorporation of auxiliary power, prediction of low power machining and predicting remaining useful life prediction.

Acknowledgements

This work was supported by the Engineering and Physical Science Research Council (EPSRC) from funding for the project "AIIM"[grant reference EP/K014137/1].

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